เลขที่นั่งสอบ

College of Industrial Technology
King Mongkut's University of Technology North Bangkok

Final Examination of Semester 2 **Year:** 2013
Subject: 341152 Electric Circuits II **Section:** 5-6
Date: 14 March 2014 **Time:** 10.00-12.00

Name:...ID:...Class:.......................

Instructions:

1. Cheating will result in failure of all classes registered for the current semester. Students who are caught on cheating will also be denied for registering in the following semester.
2. No documents are allowed to be taken out of the examination room.
3. Text books and dictionary are **NOT** allowed.
4. Calculator is allowed.
5. Electronic communication devices are **NOT** allowed in the examination room.
6. The examination has 12 pages (including this page), 3 parts (54 questions) and a total score of 95 points.
7. In part 1 and 2, enter the correct answer in the answer sheet on the LAST PAGE.
8. In part 3, show all the steps of your solution with answers in the provided space underneath each problem.

Part 1 (25 points) True/False

Instruction: Following questions, mark "X" for the correct answer, "True" or "False" in the provided answer sheet (see LAST PAGE).

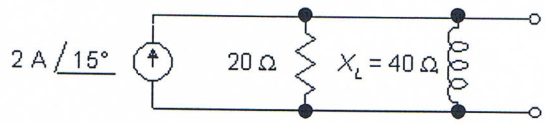
---

1. Unit of ω is in radian per second.
2. Hz is referred to frequency.
3. The negative impedance implies to inductive reactance.
4. By increasing frequency, capacitive reactance is increased.
5. In writing equations for the nodal analysis of a network, the Kirchhoff's Current Law is applied at each node.
6. The unit of average power is in VA.
7. If a load on a circuit is purely capacitive, the voltage and current are in phase.
8. The power delivered to a resistor is in the form of complex power.
9. Apparent power is equal to real power in resistive load.
10. Reactive power is positive in case of capacitive load.
11. The advantage of an autotransformer over a step-up transformer is that it provides better isolation.
12. The primary and secondary voltages of a transformer are related by the direct ratios of the primary and secondary turns.
13. In a Y-connected generator, the phase angle between any line voltage and the nearest phase voltage is 120°.
14. In a Y-connected load, the 220 V is available from line-to-neutral and the 380 V is available from line-to-line.
15. In AC networks, inductive reactance would have low reactance at high frequencies.
16. The lower the current flow for a given voltage, the smaller the value for impedance.
17. Admittance is the inverse of impedance.
18. Thevenin's theorem for AC circuits states that any 2-terminal linear AC network can be replaced by an equivalent circuit consisting of a voltage source and impedance in parallel.
19. In order to use the Superposition Theorem, there must be at least two energy sources in the network.
20. Current and voltage are out of phase in series resonant circuits.
21. For capacitor in AC circuit, the power factor is considered to be leading.
22. In resonant circuit, reactance is eliminated.
23. Complex conjugate refers to the polar form with the opposite sign of angle.
24. Power maximum transfer from DC circuit is exactly the same as those from AC circuit.
25. Capacitor can be used to increase the power factor to unity.

Part 2 (50 points) Multiple choice

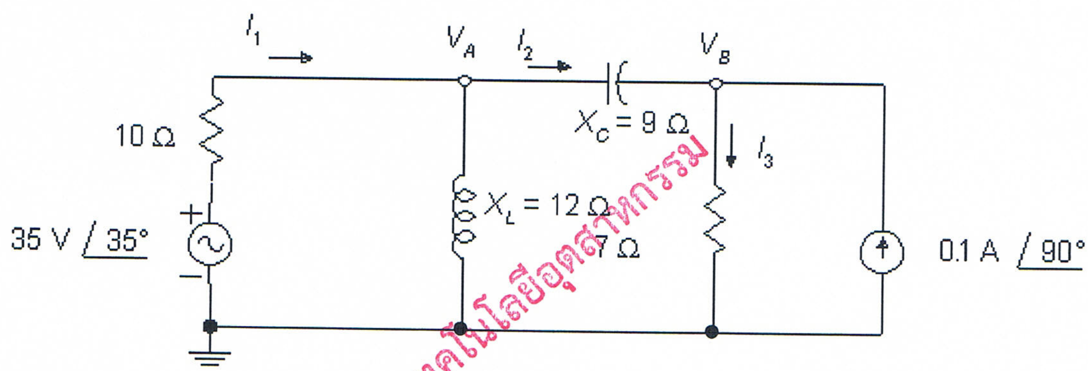
Instruction: Mark the correct answer for the following questions in the provided answer sheet (see LAST PAGE).

1. Convert the current source to a voltage source. What is the angle of the voltage?



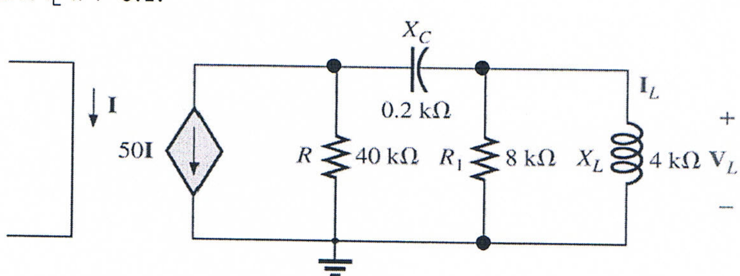
(a) 41.6° (b) 27.1° (c) 25.5° (d) 24.3°

2. Find the angle of V_B.



(a) 126.9° (b) 95.9° (c) 91.9° (d) 129.8°

3. Find the current I_L if I=0.1.

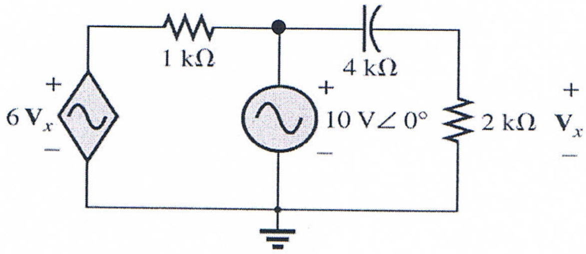


(a) 4.291 I ∠-149.31° (b) 42.91 A ∠-149.31°
(c) 4.291 A ∠149.31° (d) 42.91 A ∠149.31°

4. Which of the followings is the power factor?

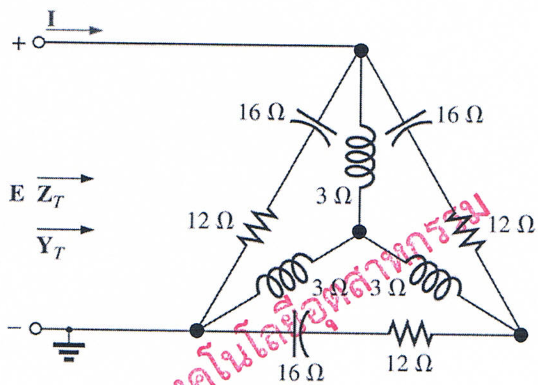
(a) P/S (b) Q/S (c) S/P (d) S/Q

5. Find the current through the 2 kΩ resistor.



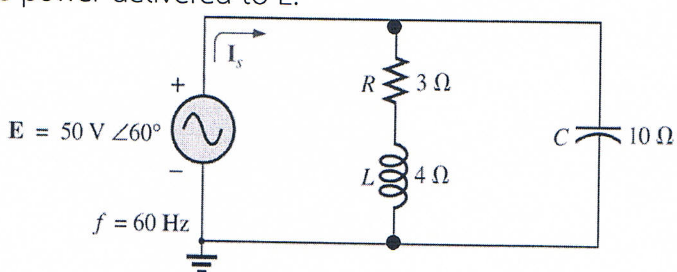
(a) 1.67 mA ∠0°
(b) 16.7 mA ∠0°
(c) 1.67 A ∠0°
(d) 16.7 A ∠0°

6. Determine the current I



(a) 11.57 mA ∠−67.13°
(b) 11.57 A ∠−67.13°
(c) 11.57 mA ∠67.13°
(d) 11.57 A ∠67.13°

7. Find the average power delivered to L.



(a) 0 W
(b) 5 W
(c) 10 W
(d) 20 W

8. From problem 7, find the reactive power delivered to L.

(a) 400 VAR
(b) 200 VAR
(c) 100 VAR
(d) 0 VAR

9. From problem 7, find the apparent power of at the capacitor.

(a) 450 VA (b) 335.41 VA (c) 300 VA (d) 250 VA

10. From the loads of 10 kW (resistive), 20 kVAR (inductive), and 10 kVAR (capacitive), find the total kilovolt-amperes.

(a) 40 kVA (b) 30 kVA (c) 14.14 kVA (d) 7.07 kVA

11. In previous problem, determine the power factor.

(a) 0.5 (b) 0.707 (c) 0.866 (d) 0.9

12. The phase voltage of a Y-connected three-phase generator is 220 V. What is the line voltage?

(a) 110 V (b) 155 V (c) 220 V (d) 380 V

13. The phase current of a Y-connected three-phase generator is 8 A. What is the line current?

(a) 4.6 A (b) 8 A (c) 11.3 A (d) 13.9 A

14. If the transformation ratio is greater than 1, what is the transformer called?

(a) step down (b) secondary (c) primary (d) step up

15. If the transformation ratio is 2.4, and the secondary voltage is 12 VDC, what is the primary voltage?

(a) 28.8 V (b) 5 V (c) 0 V (d) 0.2 V

16. What transformation ratio is needed for a transformer on a power pole to convert 4,000 VAC to 120 VAC?

(a) 0.03 (b) 5.8 (c) 10 (d) 33.3

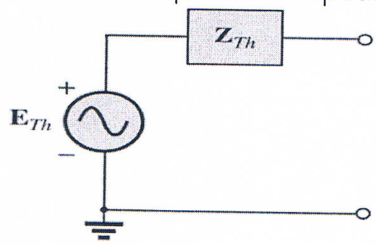
17. When using the superposition theorem in AC network analysis, which one of the following statements is false?

(a) A portion of the network produced by each sources can be added.
(b) Current sources are replaced by open circuits.
(c) Voltage sources are replaced by short circuits.
(d) All impedances are replaced by the complex conjugates.

18. If Z_{th} = 10 Ω − j10 Ω, what must the load impedance be for maximum power transfer to occur?

(a) 10 + j10 Ω (b) 10 − j10 Ω (c) −10 − j10 Ω (d) −10 + j10 Ω

19. From figure below, what is the Norton equivalent impedance Z_N?

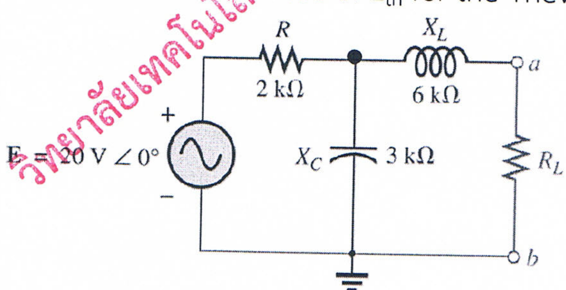


(a) Z_{th} (b) $I_{th}Z_{th}$ (c) I_{th}/Z_{th} (d) $Z_{th}/2$

20. From the problem 19, what is the Norton equivalent current I_N?

(a) Z_{th} (b) $I_{th}Z_{th}$ (c) I_{th}/Z_{th} (d) $Z_{th}/2$

21. From figure below, what should be the value of E_{th} for the Thevenin equivalent circuit?



(a) 1.66 V ∠−33.69° (b) 16.64 V ∠−33.69°

(c) 1.66 V ∠33.69° (d) 16.64 V ∠33.69°

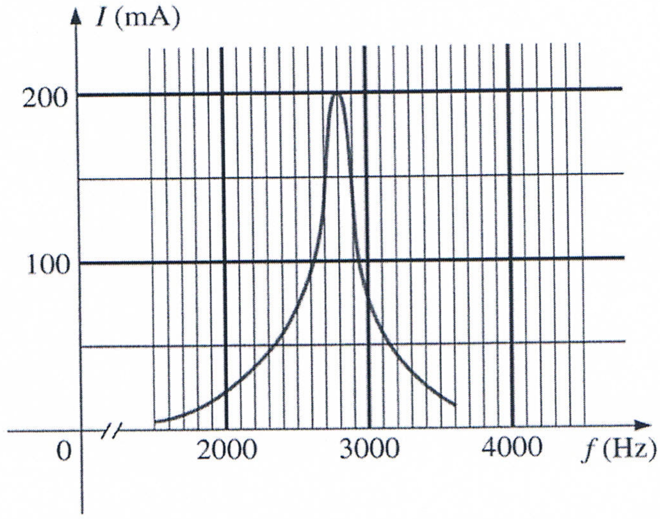
22. A network consists of E = 50 V∠20° and Z = 100 Ω∠30° connected in series. If this voltage source is converted to a current source, what is the value of the parallel impedance element Z?

(a) 2.0 Ω∠10° (b) 0.01 Ω∠−30° (c) 100 Ω∠30° (d) 0.5 Ω∠−10°

23. What is the resonant frequency of a series circuit consisting of a 100 pF capacitor, a 10 k Ω resistor, and a 1 mH inductor?

(a) 503 kHz (b) 159 Hz (c) 3.16 MHz (d) 1.58 MHz

24. The curve below describes a series resonant circuit. What is the power delivered to the load if the load looks like a 5 Ω resistance at resonance?



(a) 2 W (b) 2000 W (c) 0.2 W (d) 200 W

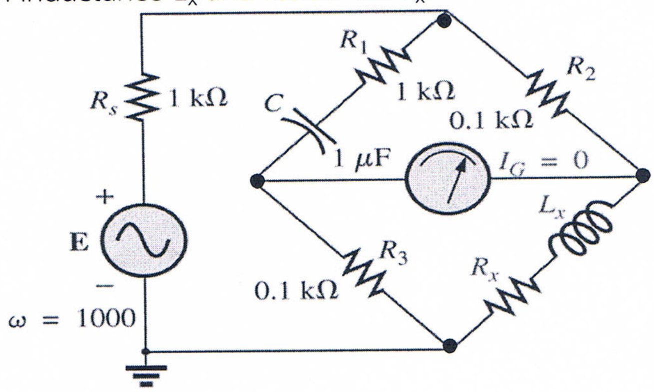
25. At 60 Hz, a series LCR circuit contains reactances X_L = 20 Ω and X_C = 24.4 Ω and resistance = 24.4 Ω. What is the impedance at resonant frequency?

(a) 26.4 Ω (b) 24.4 Ω (c) 10 Ω (d) 20 Ω

Part 3 (20 points)

Instruction: Show the mathematical expression and answers of the following problems.

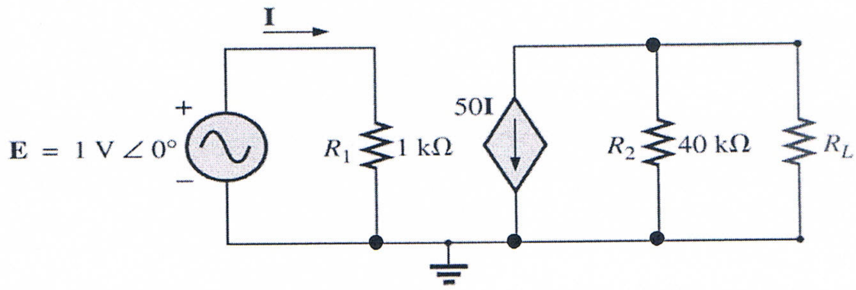
1. Find the unknown inductance L_x and resistance R_x.



2. From the network below, find

   (a) Load impedance R_L for the network below for maximum power to the load.

   (b) Maximum power to the load.



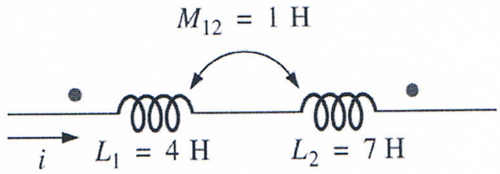
3. The loading of a factory on a 380 V, 50 Hz system includes:

- 20 kW heating (0.7 power factor)
- 10 kW motors (0.9 lagging power factor)
- 1 kW lighting (0.95 lagging power factor)

(a) Determine the capacitor required to raise the power factor to unity.

(b) Determine the capacitor required to raise the power factor to 0.98.

4. Determine the total inductance of the series coils in the figure below:



Instructors: Asst. Prof. Dr. Poolsak Koseeyaporn
and Asst. Prof. Dr. Titipong Lertwiriyaprapa

Answer Sheet

Name:________________________________ID:________________________

Subject: 341152 Electric Circuits II

Part 1

	True	False
1.		
2.		
3.		
4.		
5.		
6.		
7.		
8.		
9.		
10.		
11.		
12.		
13.		
14.		
15.		
16.		
17.		
18.		
19.		
20.		
21.		
22.		
23.		
24.		
25.		

Part 2

	a	b	c	d
1.				
2.				
3.				
4.				
5.				
6.				
7.				
8.				
9.				
10.				
11.				
12.				
13.				
14.				
15.				
16.				
17.				
18.				
19.				
20.				
21.				
22.				
23.				
24.				
25.				